Python & Tools Cheat Sheet

Week 5: Terminal, Git, and File I/O Mastery

1. The Terminal (Command Line)

COMMAND	ACTION	EXAMPLE
`ls`	List all files in current folder	`ls`
`cd`	Change directory (move to folder)	`cd projects`
`mkdir`	Make a new directory (folder)	`mkdir week5_lab`
`touch`	Create a new empty file	`touch main.py`
`rm`	Remove/Delete a file	`rm old_code.py`
`python`	Run a python script	`python main.py`

> ***Tip:*** *Use the* `Tab` *key to auto-complete file and folder names!*

2. Git & GitHub Workflow

COMMAND	WHAT IT DOES
`git init`	Turn the current folder into a Git repository.
`git add .`	Stage all changes (prepare them to be saved).
`git commit -m "..."`	Save your changes locally with a descriptive message.
`git push`	Upload your local saves to your GitHub account.
`git status`	Check which files are changed or staged.

3. Python File I/O (Read & Write)

Always use the `with` keyword to ensure files close properly.

MODE	CODE SNIPPET
Write	`with open("file.txt", "w") as f:` `  f.write("Hello!")`
Read	`with open("file.txt", "r") as f:` `  content = f.read()`

Append	`with open("file.txt", "a") as f:` `    f.write("\nNew Line")`

4. Modularity & Imports

To use functions from `tools.py` inside `main.py`:

```
# In main.py
import tools
tools.my_function()
```

To turn a folder into a **Package**, add a blank file named `__init__.py` inside it.